

Introducing NSSR Research Pathways

Discover. Connect. Research.

NSSR Research Pathways is a recurring flagship initiative designed to help students explore different research fields, connect with researchers and fellow students, develop research ideas, and take meaningful steps toward pursuing research.

Each month, **NSSR will spotlight one specific research area** — for example:

- **NSSR Research Pathways: Biophysics**
- **NSSR Research Pathways: Plasma Physics**
- **NSSR Research Pathways: Astrophysics**
- **NSSR Research Pathways: Computational Physics**

The featured field will be promoted throughout the entire month through NSSR's social media platforms, educational resources, discussions, and research-related activities.

How It Works

Summary: At the start of the month, we will start sharing about the designated research area; then we will organize an online event along with that and announce the research interest forms; lastly, we will organize the in-person research meetup at the end of the month.

1. Explore the Field

Each month, NSSR will select one research area and dedicate the month to exploring it. Throughout the month, students will be introduced to the fundamentals, current research directions, important research questions, career pathways, and opportunities within the field.

NSSR will share relevant **educational posts, research articles, introductory materials, videos, reading resources, and other useful content** to help students develop a basic understanding of the field.

2. Online Research Session

During the month, NSSR will organize **one online session** with a researcher, professor, or expert working in the featured field.

The session may cover:

- Introduction to the research field
- Current research areas and open problems

- How students can begin research in the field
- Required skills and background
- Research and career opportunities
- Experiences and advice from researchers

The purpose is not simply to give another conventional talk, but to help participants understand **how they can actually enter and explore the field.**

3. Research Interest Submission

After engaging with the month's activities, students interested in pursuing research in the featured field will be invited to fill out a **Research Interest Form.**

Students can submit either:

Individual Submission:

A student can submit their own research interest, question, or idea.

Team Submission:

If students have already formed a research team, **one member can submit the form on behalf of the entire team**, including the names and relevant information of the team members.

Individual Without a Team:

Students who do not have a team can still apply individually. They will have opportunities during the program to meet other students with similar interests and potentially form research teams or collaborations.

Depending on the stage of their idea, participants may submit either:

- A **short video explaining their research interest or idea**, or
- A **research proposal/concept note** for a more developed idea.

The form will help NSSR understand the participant's background, research interests, motivation, idea, and preferred area of research.

4. Offline Research Meetup

One **offline research meetup** will be organized during the month in collaboration with a partner institution.

The meetup will bring together students, researchers, professors, mentors, and other individuals interested in the featured research area.

Participants will have the opportunity to:

- Meet researchers and professors

- Discuss research ideas
- Ask questions directly
- Meet students with similar interests
- Find potential research collaborators
- Form research teams
- Exchange knowledge and resources
- Explore possible research opportunities

Participation in the meetup will not be limited only to students selected for direct mentorship.

Everyone participating in the Research Pathways initiative can benefit from the research community and networking opportunities.

5. Ten-Minute Research Idea Presentations

As part of the offline meetup, selected participants or teams will have the opportunity to give a **10-minute presentation** about their research idea, research question, or area of interest.

These presentations will allow participants to:

- Clearly communicate their research ideas
- Receive feedback from researchers and peers
- Improve and refine their ideas
- Find students interested in collaborating
- Connect with potential mentors
- Discover possible research directions

These presentations can also create opportunities for participants to find collaborators who may be interested in joining their research team.

6. Mentorship & Research Opportunities

After reviewing the Research Interest Forms and submissions, NSSR will identify participants whose interests and ideas are suitable for available mentorship opportunities.

Since mentor availability is limited, **direct mentorship will be offered to a limited number of selected participants or teams** based on factors such as:

- Research interest and motivation
- Clarity and potential of the research idea
- Background and preparedness
- Alignment with available mentors
- Feasibility of pursuing the proposed research
- Availability and capacity of mentors

Selected participants or teams may then be connected with suitable **researchers, professors, or supervisors** for further guidance.

Where appropriate and feasible, participants may continue developing their research ideas and potentially carry out research under the guidance of a suitable mentor.

7. Research Networking & Collaboration

NSSR Research Pathways is **not only a mentorship program**.

Even if a participant is not selected for direct mentorship, they will still have the opportunity to become part of a research-focused community.

Participants can:

- Connect with other students
- Find potential research partners
- Form new research teams
- Exchange ideas
- Discuss research opportunities
- Learn from other participants' presentations
- Interact with researchers and professors
- Build connections for future research opportunities

In this way, NSSR Research Pathways creates opportunities not only for **student–mentor connections**, but also for **student–student and student–researcher collaborations**.

The Research Pathway

The initiative aims to create a clear journey from initial curiosity to actual research:

Discover a Field → Learn → Develop an Idea → Submit Your Interest → Present & Receive Feedback → Connect → Collaborate → Access Mentorship → Pursue Research

Not every participant will necessarily reach the mentorship stage immediately, but every participant will have an opportunity to learn, connect, and become part of the research community.

The Vision

Through **NSSR Research Pathways**, NSSR aims to move beyond the traditional model of organizing talks and seminars.

The goal is to build a **continuous research ecosystem** where students can discover fields they are passionate about, learn from experts, develop and communicate their own ideas, find collaborators, connect with mentors, and eventually take their first steps toward conducting meaningful research.